

Methylation Prediction Results

Date & Time: 2026-06-29 16:29:15

Classification Report Of Random Forest:

Metric	Non-Methylated	Methylated
Precision	0.8970	0.7377
Recall	0.9086	0.7113
F1-Score	0.9028	0.7242
Support	930	336
Accuracy	85.62%	
MCC	0.6273	
AUC	0.8999	

Classification Report Of XGBoost:

Metric	Score
Accuracy	84.12%
MCC	0.5917
AUC	0.8958

Classification Report Of CatBoost:

Metric	Score
Accuracy	82.54%
MCC	0.5536
AUC	0.8791

Classification Report Of LightGBM:

Metric	Score
Accuracy	84.83%
MCC	0.6068
AUC	0.902

Classification Report Of SVM:

Metric	Score
Accuracy	69.83%
MCC	0.3304
AUC	0.7285

Classification Report Of ESM + RF:

Metric	Score
Accuracy	84.52%
MCC	0.6045
AUC	0.8922

Classification Report Of ESM + SVM:

Metric	Score
Accuracy	75.28%
MCC	0.4734
AUC	0.8315

Classification Report Of Hyperparameter Tuned RF:

Metric	Score
Accuracy	84.99%
MCC	0.6088
AUC	0.9043

Classification Report Of Neural Network:

Metric	Score
Accuracy	78.12%
MCC	0.5067
AUC	0.8462

Classification Report Of Naive Bayes:

Metric	Score
Accuracy	69.35%
MCC	0.3905
AUC	0.7364

Classification Report Of KNN:

Metric	Score
Accuracy	70.77%
MCC	0.412
AUC	0.7938

Classification Report Of ESM + KNN:

Metric	Score
Accuracy	74.96%
MCC	0.4875
AUC	0.838

Classification Report Of ESM + Tuned RF (GridSearchCV):

Metric	Score
Accuracy	83.18%
MCC	0.5714
AUC	0.8845

Classification Report Of ESM + Feature Selection:

Metric	Score
Accuracy	85.15%
MCC	0.6177
AUC	0.9012

Classification Report Of Ensemble (RF+XGB+LGBM):

Metric	Score
Accuracy	85.94%
MCC	0.6361
AUC	0.9075

Model Comparison Table

Model	Accuracy	MCC	AUC
Ensemble RF+XGB+LGBM	85.94%	0.6361	0.9075
Random Forest	85.62%	0.6273	0.8999
Tuned RF	84.99%	0.6088	0.9043
Feature Selection	85.15%	0.6177	0.9012
LightGBM	84.83%	0.6068	0.902
ESM + RF	84.52%	0.6045	0.8922
XGBoost	84.12%	0.5917	0.8958
CatBoost	82.54%	0.5536	0.8791
Neural Network	78.75%	0.5317	0.8487
ESM + SVM	75.28%	0.4734	0.8315
KNN + ESM	74.96%	0.4875	0.838
KNN	70.77%	0.412	0.7938
SVM	69.83%	0.3304	0.7285
Naive Bayes	69.35%	0.3905	0.7364

Model Comparison – Methylation Prediction

